PROJECT SUMMARY

Sales Optimisation & Revenue Growth Analysis for a Multistore Supermarket Chain

1. PROJECT OVERVIEW

Supermarket retail businesses operate in a high-volume, low-margin environment where even small fluctuations in customer behaviour can significantly impact overall revenue performance. Identifying early signals of sales decline and understanding their structural causes is critical for sustainable operations.

This project analyses structured transactional retail data to investigate declining sales performance across a one-year period. The objective is to determine whether the decline is driven by pricing changes, transaction volume contraction, customer behaviour shifts, product imbalance, or operational factors.

The project integrates Python-based Exploratory Data Analysis (EDA), Tableau performance storytelling, and Power BI executive dashboards to develop a comprehensive analytical and decision-support framework.

2. PROBLEM STATEMENT

Supermarket businesses frequently face fluctuations in sales performance due to:

- Seasonal demand variation
- Changing customer behaviour
- Product mix imbalance
- Operational inefficiencies
- Segment dependency risks

However, identifying the exact structural driver behind sales decline is often unclear.

Core Objective:
To analyse supermarket transactional data and determine the primary factors contributing to declining sales performance using structured data analytics techniques.

3. DATASET OVERVIEW

The dataset used for this study is a publicly available supermarket retail dataset obtained from Kaggle. It contains structured transaction-level information covering one full calendar year.

Dataset Information Table

Attribute	Details
Total Transactions	1,000
Total Features	21 (including 4 derived features)
Time Period	January – December 2019
Branches	A, B, C
Cities	Yangon, Mandalay, Naypyitaw
Product Lines	6
Customer Types	Member, Normal
Payment Methods	Cash, Credit Card, E-wallet

Column Classification

Numerical Columns:
Quantity, Unit Price, Tax %, Total, COGS, Gross Income, Gross Margin %, Rating

Categorical Columns:
Branch, City, Customer Type, Gender, Product Line, Payment, Day Name, Invoice ID

Date & Time Columns:
Date, Time, Hour, Day, Month

Derived Features (Feature Engineering):
Hour, Day, Month, Day Name

This dataset structure enabled multi-dimensional analysis across time, customer behavior, product performance, and branch operations.

4. METHODOLOGY

The analysis followed a structured multi-stage framework:

1. Base Plot Identification

Monthly Revenue was selected as the base performance indicator to anchor the exploratory process.

2. Exploratory Data Analysis (Python)

- Univariate analysis to understand distribution patterns
- Bivariate analysis to examine relationships between variables
- Correlation analysis to identify strong associations
- Hypothesis testing for statistical validation

3. Revenue Decomposition (Tableau)

Revenue was decomposed into:

Revenue = Price × Quantity

This helped distinguish pricing effects from volume effects.

4. Performance Classification

Monthly performance was categorised into:

- High
- Moderate
- Low

This allowed the detection of structural shifts in business regimes.

5. Dashboard Development (Power BI)

Interactive dashboards were developed to:

- Monitor branch contribution
- Track product distribution
- Analyse customer behaviour
- Evaluate operational timing
- Support executive-level decision making

5. RESULTS & KEY FINDINGS

The analysis revealed the following insights:

• Revenue decline was primarily driven by **reduction in transaction volume**.
• **Average selling price** remained relatively **stable** throughout the period.
• **Customer participation** reduced during **lower performance months**.
• **Performance regime** shifted from **high** to **lower levels** without full recovery.
• **Branch contribution** remained **balanced**, indicating **systemic decline** rather than localized issues
• **Evening hours** showed **peak sales activity**, with slight **service strain reflected in ratings**.

Performance Regime Thresholds

- **High Performance:** ~996 units (monthly average)
- **Moderate Performance**: ~301 units
- **Low Performance**: ~269 units

This classification confirmed **structural volume contraction** as **the dominant driver of sales decline**.

6. STRATEGIC RECOMMENDATIONS

Based on the analytical findings, the following strategies are proposed:

• Monitor monthly volume thresholds for early detection of decline.
• Strengthen engagement for high-contribution customer segments.
• Improve operational efficiency during peak hours.
• Maintain minimum performance benchmark levels.
• Implement targeted retention strategies instead of generalized discounting.

7. CONCLUSION

This project demonstrates that declining supermarket sales performance is primarily driven by structural volume contraction rather than pricing instability.

Through structured EDA, performance regime classification, and executive dashboard development, a comprehensive decision-support framework was established.

The framework enables early detection of performance decline and supports data-driven strategic intervention in retail environments.